

Linear Algebra 1A Cheat Sheet

Convention: a matrix of size $r \times c$ has r rows and c columns.

1. Fields

A field $\mathbb{F}$ is a set with operations $(+, \cdot)$ satisfying: closure, associativity, commutativity, identities $(0, 1)$, inverses, and distributivity.

Characteristic.

- $\text{char}(\mathbb{F}) :=$ the least $n \geq 1$ with $\overbrace{1 + \cdots + 1}^{n \text{ times}} = 0$, and $\text{char}(\mathbb{F}) := 0$ if no such n exists
- $\text{char}(\mathbb{F})$ is always 0 or prime
- $\forall \mathbb{F} : \text{char}(\mathbb{F}) = p > 0 \Rightarrow \mathbb{Z}_p$ is a subfield of $\mathbb{F}$

Finite fields.

- $|\mathbb{F}| < \infty \Rightarrow |\mathbb{F}| = p^r$, where $p = \text{char}(\mathbb{F})$ and $r \in \mathbb{N}$
- $\forall (p \in \mathbb{P}) \cdot \forall (r \in \mathbb{N}) \cdot \exists (\mathbb{F} \text{ field}) : |\mathbb{F}| = p^r$

2. Matrices

$$A_{m \times p} B_{p \times n} = (AB)_{m \times n}$$

- $(AB)_{ij} := \sum_{k=1}^p a_{ik} b_{kj}$
- B with columns $b_1, \dots, b_n$: $AB = (Ab_1, \dots, Ab_n)$
- A with rows $a_1, \dots, a_m$: row i of AB is $a_i B$
- $AB = 0$ does not imply $A = 0$ or $B = 0$
- If $A^T = A$ and $B^T = B$, then AB is symmetric $\iff AB = BA$
- φ an elementary row operation, $A \in \mathbb{F}^{m \times n}$: $\varphi(A) = \varphi(I_m) A$

3. Vector Spaces & Subspaces

A subset $W \subseteq V$ is a subspace if and only if:

- W is nonempty (equivalently $\vec{0} \in W$)
- Closed under addition: if $u, v \in W$ then $u + v \in W$
- Closed under scalar multiplication: if $\alpha \in \mathbb{F}$ and $u \in W$ then $\alpha u \in W$

Equivalently: $\alpha u + v \in W$ for all $u, v \in W$ and all $\alpha \in \mathbb{F}$.

- $U \cap W$ is a subspace of V
- $U \cup W$ is a subspace of $V \iff U \subseteq W$ or $W \subseteq U$
- $U + W := \{u + w : u \in U, w \in W\} = \text{span}(U \cup W)$ is a subspace of V
- Direct sum: $U \oplus W \iff U \cap W = \{\vec{0}\}$ (general case: see Direct sums, §6)
- $\text{span}(S)$ is the smallest subspace containing S
- $S_1 \subseteq S_2 \subseteq V \Rightarrow \text{span}(S_1) \subseteq \text{span}(S_2) \subseteq V$
- $W \leq V \Rightarrow \text{span}(W) = W$
- $S \subseteq W \leq V \Rightarrow \text{span}(S) \leq W$

4. Rank

- $\text{rank}(A) :=$ the number of pivots in $\text{RREF}(A)$
- $\text{rank}(A_{m \times n}) \leq \min\{m, n\}$
- $\text{rank}(A^T) = \text{rank}(A)$
- $A \sim B \Rightarrow \text{rank}(A) = \text{rank}(B)$
- $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$
- $A, B \in \mathbb{F}^{n \times n}, AB = 0 \Rightarrow \text{rank}(A) + \text{rank}(B) \leq n$

5. Linear Systems

$Ax = b$, n variables, $r = \text{rank}(A)$

- Number of pivots $= r$, free variables $= n - r$
- System is solvable $\iff b \in \text{Col}(A)$
- x_p : any one solution of $Ax = b$ (a *particular* solution)
- to find x_p : reduce $[A \mid b]$, set every free variable to 0, then
 - from RREF: read off the pivot variables
 - from row echelon form: back-substitute
- Homogeneous system: $Ax = \vec{0}$ has solution space $\text{Null}(A) = \ker(A)$, of dimension $n - r$
- $\{x : Ax = b\} = x_p + \ker(A)$

6. Bases & Dimension

If $\dim V = n$:

- Any linearly independent set has size at most n
- Any spanning set has size at least n
- $\dim V < \infty$: any linearly independent set extends to a basis of V
- $\dim V < \infty$: any spanning set contains a basis of V
- A set B with $|B| = n$ is a basis $\iff$ it is linearly independent $\iff$ it spans V
- If $W \leq V$, then $W = V \iff \dim W = \dim V$
- $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$

Direct sums. Let $M_1, \dots, M_s \leq V$ with

$M = M_1 + \cdots + M_s$. The following are equivalent:

- $M = M_1 \oplus \cdots \oplus M_s$
- $\forall a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
- $\exists a \in M$: the representation $a = a_1 + \cdots + a_s$, $a_i \in M_i$, is unique
- $\forall a_i \in M_i : a_1 + \cdots + a_s = \vec{0} \Rightarrow a_1 = \cdots = a_s = \vec{0}$
- $E^{(i)}$ a basis of $M_i \Rightarrow E^{(1)}, \dots, E^{(s)}$ together form a basis of M
- $\dim M = \dim M_1 + \cdots + \dim M_s$

Note 3 $\Rightarrow$ all: one witness with a unique decomposition suffices.

7. Coordinates

Let $B = \{v_1, \dots, v_n\}$ be a basis of V .

- $[B] := (v_1, \dots, v_n)$, the basis written as a row of its vectors
- Every $v \in V$ has a unique representation $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$
- Coordinate vector: $[v]_B := (\alpha_1, \dots, \alpha_n)^T$
- $v = [B][v]_B$
- $u_1, \dots, u_n$ linearly independent, $A \in \mathbb{F}^{n \times p}$: $(u_1, \dots, u_n)A = (\vec{0}, \dots, \vec{0}) \Rightarrow A = 0$

Change of basis. The transition matrix from basis B to basis C , written $[I]_{B \rightarrow C}$ (also $[I]_B^C$), has the $[v_i]_C$ as its columns:

- $[I]_{B \rightarrow C} := ([v_1]_C, \dots, [v_n]_C)$
- $[v]_C = [I]_{B \rightarrow C} [v]_B$
- B, C, D bases: $[I]_{B \rightarrow D} = [I]_{C \rightarrow D} [I]_{B \rightarrow C}$
- $[I]_{B \rightarrow C}$ is invertible, with $([I]_{B \rightarrow C})^{-1} = [I]_{C \rightarrow B}$
- $[C] = (w_1, \dots, w_n) : [C] = [B]P \Rightarrow P = [I]_{C \rightarrow B}$

8. Invertible Matrices

- P invertible $\Rightarrow \text{rank}(PA) = \text{rank}(AP) = \text{rank}(A)$
- $\alpha \neq 0 \Rightarrow (\alpha A)^{-1} = \alpha^{-1} A^{-1}$
- C invertible, $B = CA$: B invertible $\iff A$ invertible

Conditions Related to Invertibility

Let $A \in \mathbb{F}^{n \times n}$. Each item states its relation to “ A is invertible”:

1. $\implies A^{-1}$ is unique
2. $\implies A$ has no zero row or column
3. $\iff \text{rank}(A) = n$
4. $\iff \det(A) \neq 0$
5. $\iff A^T$ is invertible, with $(A^T)^{-1} = (A^{-1})^T$
6. $\iff A \sim I_n$
7. $\iff \text{RREF}(A) = I_n$
8. $\iff Ax = \vec{0} \Rightarrow x = \vec{0}$
9. $\iff$ For all $b \in \mathbb{F}^n$, the system $Ax = b$ has the unique solution $x = A^{-1}b$
10. $\iff \text{Row}(A) = \mathbb{F}^n$
11. $\iff \text{Col}(A) = \mathbb{F}^n$
12. $\iff$ The rows of A span $\mathbb{F}^n$
13. $\iff$ The columns of A span $\mathbb{F}^n$
14. $\iff$ The rows of A are linearly independent
15. $\iff$ The columns of A are linearly independent
16. $\iff$ The rows of A form a basis of $\mathbb{F}^n$
17. $\iff$ The columns of A form a basis of $\mathbb{F}^n$
18. $\iff \dim \text{Row}(A) = n$
19. $\iff \dim \text{Col}(A) = n$
20. $\iff A$ is a product of elementary matrices
21. $\iff \text{adj}(A)$ is invertible
22. $\iff A$ is invertible on one side (i.e. $AB = I$ or $BA = I$ for some B) — **square A only**
23. $\iff A = [I]_{B \rightarrow C}$ for some bases B, C
24. $\iff A$ is elementary (then A^{-1} is also elementary)

9. Determinants & Adjugate

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \det(B)$
- $\det(A^{-1}) = (\det A)^{-1}$
- $\det(\alpha A) = \alpha^n \det(A)$ for $A \in \mathbb{F}^{n \times n}$

Adjugate.

$A \in \mathbb{F}^{n \times n}$

- $M_{ij}(A) := A$ with row i and column j deleted
- cofactor $c_{ij} := (-1)^{i+j} \det(M_{ij}(A))$
- $\text{adj}(A)_{ij} := c_{ji} = (-1)^{i+j} \det(M_{ji}(A))$
- $\text{adj}(A)A = A \text{adj}(A) = \det(A)I_n$
- $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$
- $\text{rank}(\text{adj}(A)) = \begin{cases} n, & \text{rank}(A) = n, \\ 1, & \text{rank}(A) = n - 1, \\ 0, & \text{rank}(A) < n - 1 \end{cases}$

Cramer's rule.

- $\det(A) \neq 0 \Rightarrow Ax = b$ has the unique solution $x_k = \frac{\det(A_k)}{\det(A)}$, $k = 1, \dots, n$
- A_k is A with column k replaced by b

10. Matrix Spaces

$A \in \mathbb{F}^{m \times n}$

- $\text{Row}(A) := \text{span}(\text{rows of } A) \leq \mathbb{F}^n$
- $\text{Col}(A) := \text{span}(\text{columns of } A) \leq \mathbb{F}^m$
- $\dim \text{Row}(A) = \dim \text{Col}(A) = \text{rank}(A)$
- **ROW OPERATIONS DO NOT PRESERVE THE COLUMN SPACE**
 - $\text{Row}(A)$ is preserved
 - $\dim \text{Col}(A) = \text{rank}(A)$ is preserved, $\text{Col}(A)$ is not
- $\text{Null}(A) = \ker(A) = \{x : Ax = \vec{0}\}$
- $\dim \ker(A) = n - \text{rank}(A)$
- $\text{Row}(AB) \subseteq \text{Row}(B)$
- $\text{Col}(AB) \subseteq \text{Col}(A)$

11. Linear Maps

$T : V \rightarrow U$ is linear if for all $u, v \in V$ and $\alpha \in \mathbb{F}$:

- $T(u + v) = T(u) + T(v)$
- $T(\alpha v) = \alpha T(v)$

Equivalently: $T(\alpha u + v) = \alpha T(u) + T(v)$.

- $\{v_1, \dots, v_n\}$ a basis of V : any choice of $T(v_1), \dots, T(v_n) \in U$ determines a unique linear $T : V \rightarrow U$
- T_1, T_2 linear $\Rightarrow T_1 \circ T_2$ is linear
- $\{v_1, \dots, v_s\}$ linearly dependent $\Rightarrow \{T(v_1), \dots, T(v_s)\}$ linearly dependent
- Kernel: $\ker T := \{v \in V : T(v) = \vec{0}\}$
 - $\ker T \leq V$
- Image: $\text{Im } T := \{T(v) : v \in V\}$
 - $\text{Im } T \leq U$
- $\dim V < \infty$, $\{e_1, \dots, e_k\}$ a basis of $\ker T$ extended to a basis $\{e_1, \dots, e_n\}$ of V $\Rightarrow \{T(e_{k+1}), \dots, T(e_n)\}$ is a basis of $\text{Im } T$
- $\dim V = \dim \ker T + \dim \text{Im } T$

12. Operators & Isomorphisms

Let $T : V \rightarrow U$ be linear.

- Monomorphism := injective T
 - T_1, T_2 mono $\Rightarrow T_1 \circ T_2$ mono
- Epimorphism := surjective T
 - T_1, T_2 epi $\Rightarrow T_1 \circ T_2$ epi
- Isomorphism := bijective T ; write $V \cong U$ if one exists
 - T_1, T_2 iso $\Rightarrow T_1 \circ T_2$ iso
 - T iso $\Rightarrow T^{-1}$ iso
 - $\cong$ is an equivalence relation
 - $\dim V, \dim U < \infty$: T iso $\iff T$ maps every basis of V to a basis of $U \iff T$ maps some basis of V to a basis of U
- T injective $\iff \ker T = \{\vec{0}\}$
- T surjective $\iff \dim \text{Im } T = \dim U$
- $V \cong U \iff \dim V = \dim U$
- $\dim V = \dim U < \infty$: T mono $\iff T$ epi $\iff T$ iso
- For an operator $T : V \rightarrow V$:
 - $\ker T \subseteq \ker T^2 \subseteq \dots$, $\text{Im } T \supseteq \text{Im } T^2 \supseteq \dots$
 - $\ker T^k = \ker T^{k+1} \Rightarrow \ker T^m = \ker T^k$ for all $m \geq k$ (same for Im)
 - $\dim V < \infty \Rightarrow$ this equality occurs, by $k = \dim V$ at the latest

13. Matrix Representation

$T : V \rightarrow W$ linear

$B = \{v_1, \dots, v_n\}$ basis of V

$C = \{w_1, \dots, w_m\}$ basis of W

$[\varphi]_B := [\varphi]_{B \rightarrow B}$ for $\varphi : V \rightarrow V$

- $[T]_{B \rightarrow C} := ([T(v_1)]_C, \dots, [T(v_n)]_C) \in \mathbb{F}^{m \times n}$
- $[T]_{B \rightarrow C} [v]_B = [T(v)]_C$ for all $v \in V$
- $\text{rank}([T]_{B \rightarrow C}) = \dim \text{Im } T$
- $[T_1]_{B \rightarrow C} = [T_2]_{B \rightarrow C} \Rightarrow T_1 = T_2$
- T is invertible $\iff [T]_{B \rightarrow C}$ is invertible
- $S : W \rightarrow Z$, D basis of Z :
 $[S \circ T]_{B \rightarrow D} = [S]_{C \rightarrow D} [T]_{B \rightarrow C}$
- B' basis of V , C' basis of W :
 $[T]_{B' \rightarrow C'} = [I]_{C \rightarrow C'} [T]_{B \rightarrow C} [I]_{B' \rightarrow B}$
 - $\varphi : V \rightarrow V$, same basis both sides:
 $[\varphi]_{B'} = P^{-1} [\varphi]_B P$, where $P := [I]_{B' \rightarrow B}$

14. The Space $\mathcal{L}(V, W)$

$\mathcal{L}(V, W) := \{\varphi : V \rightarrow W \mid \varphi \text{ is linear}\}$

$\varphi, \psi \in \mathcal{L}(V, W)$, $\lambda \in \mathbb{F}$

B basis of V , $\dim V = n$; C basis of W , $\dim W = m$

- $(\varphi + \psi)(x) := \varphi(x) + \psi(x)$
- $(\lambda\varphi)(x) := \lambda(\varphi(x))$
- $\varphi + \psi$ and $\lambda\varphi$ are linear
- $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$
- $\varphi \mapsto [\varphi]_{B \rightarrow C}$ is an isomorphism $\mathcal{L}(V, W) \rightarrow \mathbb{F}^{m \times n}$
- $\dim V, \dim W < \infty$
 $\Rightarrow \dim \mathcal{L}(V, W) = \dim V \cdot \dim W = mn$

15. Dual Spaces, Functionals, and Annihilators

Linear functionals. A linear functional is a linear map $f : V \rightarrow \mathbb{F}$. It assigns a scalar to each vector and represents a linear measurement.

Dual space.

- $V^* := \mathcal{L}(V, \mathbb{F}) = \{f : V \rightarrow \mathbb{F} \mid f \text{ is linear}\}$
- V^* is a vector space
- $\dim V = n < \infty \Rightarrow \dim V^* = n$

Dual basis.

$B = \{v_1, \dots, v_n\}$ basis of V

- $B^* := \{f_1, \dots, f_n\}$, the unique functionals with $f_i(v_j) = \delta_{ij}$
- B^* is a basis of V^* , determined by B
- $\varphi = \sum_i \varphi(v_i) f_i$ for all $\varphi \in V^*$
- $f_i(v) = f_i(\sum_j \alpha_j v_j) = \alpha_i$, the i -th coordinate of $[v]_B$

Annihilator (MEAPES).

$S \subseteq V$

- $S^\circ := \{f \in V^* : f(s) = 0 \text{ for all } s \in S\}$
- S° is a subspace of V^*
- $S_1 \subseteq S_2 \Rightarrow S_1^\circ \supseteq S_2^\circ$
- $S^\circ = (\text{span } S)^\circ$
- $\{S_i\}_{i \in I}$ subsets of V : $(\bigcup_i S_i)^\circ = \bigcap_i S_i^\circ$
- $W \leq V$, $\dim V < \infty$: $\dim W + \dim W^\circ = \dim V$

Pre-annihilator (TAT MERHAV HA EFASIM).

$G \subseteq V^*$

- $G_\circ := \{v \in V : f(v) = 0 \text{ for all } f \in G\} = \bigcap_{f \in G} \ker f$
- $G_\circ$ is a subspace of V
- $G_1 \subseteq G_2 \Rightarrow (G_1)_\circ \supseteq (G_2)_\circ$
- $G_\circ = (\text{span } G)_\circ$
- $\{G_i\}_{i \in I}$ subsets of V^* : $(\bigcup_i G_i)_\circ = \bigcap_i (G_i)_\circ$
- $N \leq V^*$, $\dim V < \infty$: $\dim N + \dim N_\circ = \dim V$
- $\emptyset_\circ = V$

Difference between MERHAV HA EFES and MEAPES.

- MERHAV HA EFES: vectors in V mapped to zero by a linear map
- MEAPES: functionals in V^* that vanish on a subset

Second dual space.

- $V^{**} := (V^*)^*$
- canonical map $\Phi : V \rightarrow V^{**}$, $\Phi(v)(f) := f(v)$
◦ *canonical*: defined with no choice of basis
- Φ is linear
- $\dim V < \infty \Rightarrow \Phi$ is an isomorphism
- $G \subseteq V^*$, $\dim V < \infty$: $\Phi(G_\circ) = G^\circ$ (in V^{**})
◦ $\dim G_\circ = \dim G^\circ$

Dual (transpose) map.

$T : V \rightarrow U$

- $T^* : U^* \rightarrow V^*$, $T^*(f) := f \circ T$
- $\ker T^* = (\text{Im } T)^\circ$
- $\text{Im } T^* = (\ker T)^\circ$
- **matrix form:** $[T^*] = [T]^T$

16. Vandermonde Matrix

$x_1, \dots, x_n \in \mathbb{F}$; the *Vandermonde matrix* is

$$V(x_1, \dots, x_n) := \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}$$

- $\det V(x_1, \dots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$
- $V(x_1, \dots, x_n)$ is invertible $\iff x_1, \dots, x_n$ are pairwise distinct
- $x_i = x_j$ for some $i \neq j \Rightarrow$ two rows are equal $\Rightarrow \det V = 0$
- all x_i distinct $\Rightarrow \text{rank}(V) = n$
- appears naturally in polynomial interpolation

17. Permutations

- a *permutation* of $\{1, \dots, n\}$ is a bijection $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$
- S_n : the set of all permutations

Inversions.

- an inversion of σ is a pair (i, j) with $i < j$ and $\sigma(i) > \sigma(j)$
- $\text{inv}(\sigma)$: the number of inversions of σ , sometimes denoted $S(\sigma)$

Sign of a permutation.

- $\text{sgn}(\sigma) := (-1)^{\text{inv}(\sigma)}$
- σ is *even* if $\text{sgn}(\sigma) = 1$
- σ is *odd* if $\text{sgn}(\sigma) = -1$

Properties.

- $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$
- A transposition changes the sign
- $|S_n| = n!$

Determinant (Leibniz formula).

$$A = (a_{ij}) \in \mathbb{F}^{n \times n}$$

- $\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}$
- two rows (or columns) equal $\Rightarrow$ every term cancels $\Rightarrow \det(A) = 0$
- swapping two rows multiplies $\det(A)$ by -1
- linearity of the determinant follows from the formula

18. Block Matrices

Definition.

A *block matrix* is a matrix partitioned into submatrices of compatible sizes

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$$

Block operations.

Block sizes agree $\Rightarrow$ blocks behave like entries

- addition and scalar multiplication are blockwise
- multiplication follows the ordinary rule

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{pmatrix}$$

Block diagonal matrices.

$$A = \begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}$$

- $\det(A) = \det(A_1)\det(A_2)$

- A invertible $\iff A_1, A_2$ invertible

Block triangular matrices.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}$$

- $\det(A) = \det(A_{11})\det(A_{22})$
- $\text{rank}(A) = \text{rank}(A_{11}) + \text{rank}(A_{22})$

Schur complement.

$$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad A_{11} \text{ invertible}$$

- *Schur complement* of A_{11} in A :
 $S := A_{22} - A_{21}A_{11}^{-1}A_{12}$
- A invertible $\iff A_{11}$ and S invertible
- $\det(A) = \det(A_{11})\det(S)$

Block inverse formula.

A_{11}, S invertible:

$$A^{-1} = \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}S^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}S^{-1} \\ -S^{-1}A_{21}A_{11}^{-1} & S^{-1} \end{pmatrix}.$$

Solving block systems.

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b \\ c \end{pmatrix}$$

1. $A_{11}x = b - A_{12}y$
2. $(A_{22} - A_{21}A_{11}^{-1}A_{12})y = c - A_{21}A_{11}^{-1}b$

Implementation tips.

- Use block structure to reduce computation
- Common in Gaussian elimination, LU decomposition, and numerical methods
- Essential for large sparse systems and matrix factorizations